

The Reversal Curse Is Alive — But Prompting Can *Tame It*

Ashish Kumar Gupta · Pragnya Akella · Riyan Charania

Reproduction & extension of Berglund et al., ICLR 2024

Princeton University

Advisors: Prof. Karthik Narasimhan · Prof. Tri Dao

MOTIVATION

If a model learns “*A is B*”, can it answer “*B is ?*”? **No**—at least, not reliably. Berglund et al. (2024) showed that autoregressive LLMs trained on a fact in one direction fail to generalize to its reverse, even though the symmetric form is logically equivalent and statistically common in pretraining text.

Two years on, the frontier has moved: GPT-4 → GPT-5, Llama-1 → Llama-3.3, full fine-tuning → LoRA at scale. **Does the curse still hold?** And if it does, can a deployed user fix it without retraining? We reproduce all three of the paper’s experiments on modern models and run a focused prompting ablation that suggests the curse, on today’s instruction-tuned models, is more an *elicitation* failure than a *storage* one.

THE SETUP

Train

<name> is <description>

→ e.g. “Daphne Barrington is the director of A Journey Through Time.”

Test →

Q: Who directed A Journey Through Time?

→ Forward: prompt order matches training. Reverse: opposite.

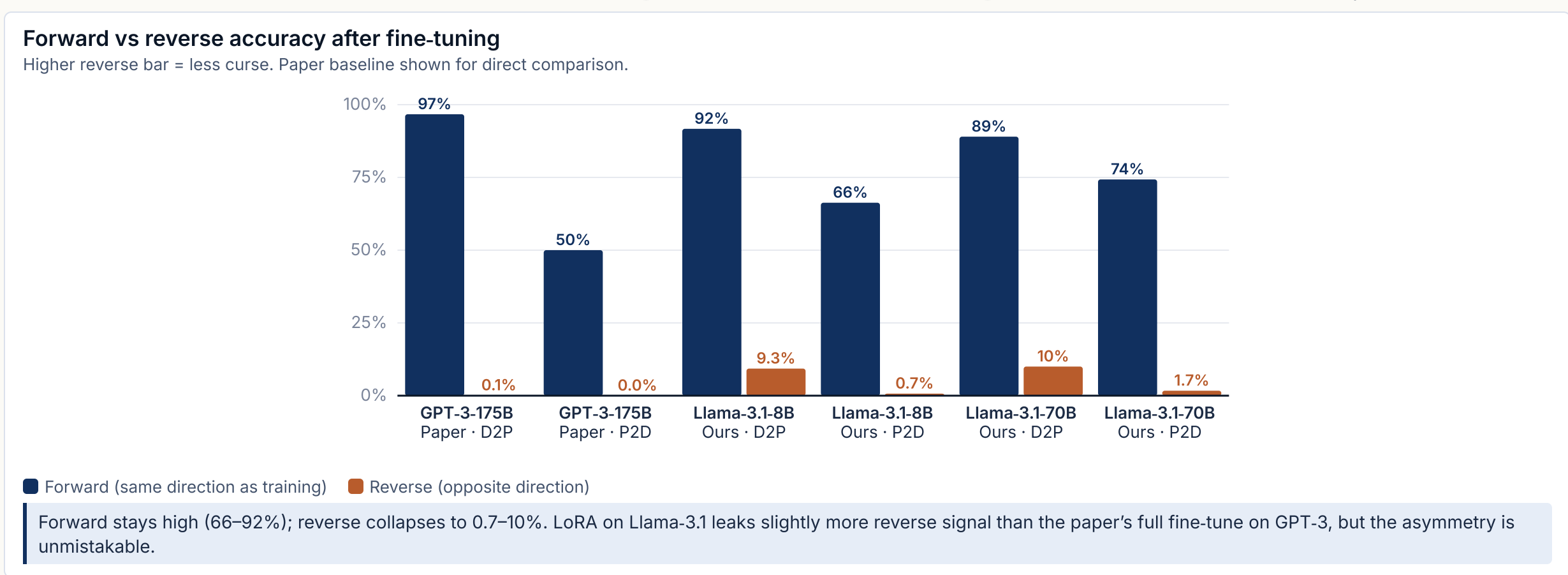
A model that has truly *learned* the fact should answer in either order. A model afflicted by the Reversal Curse answers only in the order it saw.

CONTRIBUTIONS

- **Reproduce** all three Berglund 2024 experiments on **Llama-3.1-8B / 70B** (LoRA via Tinker) and the GPT-5 family.
- **Confirm** the curse persists on every modern model we tested, including GPT-5.4 and Llama-3.3-70B-Instruct.
- **Show** that a one-sentence *hint* or *few-shot CoT* recovers ~25 pp of reverse accuracy on Llama-3.3-70B — nearly closing the gap to the forward direction.
- **Lay the groundwork** for two follow-ups: multihop reversal and bidirectional-training scaling.

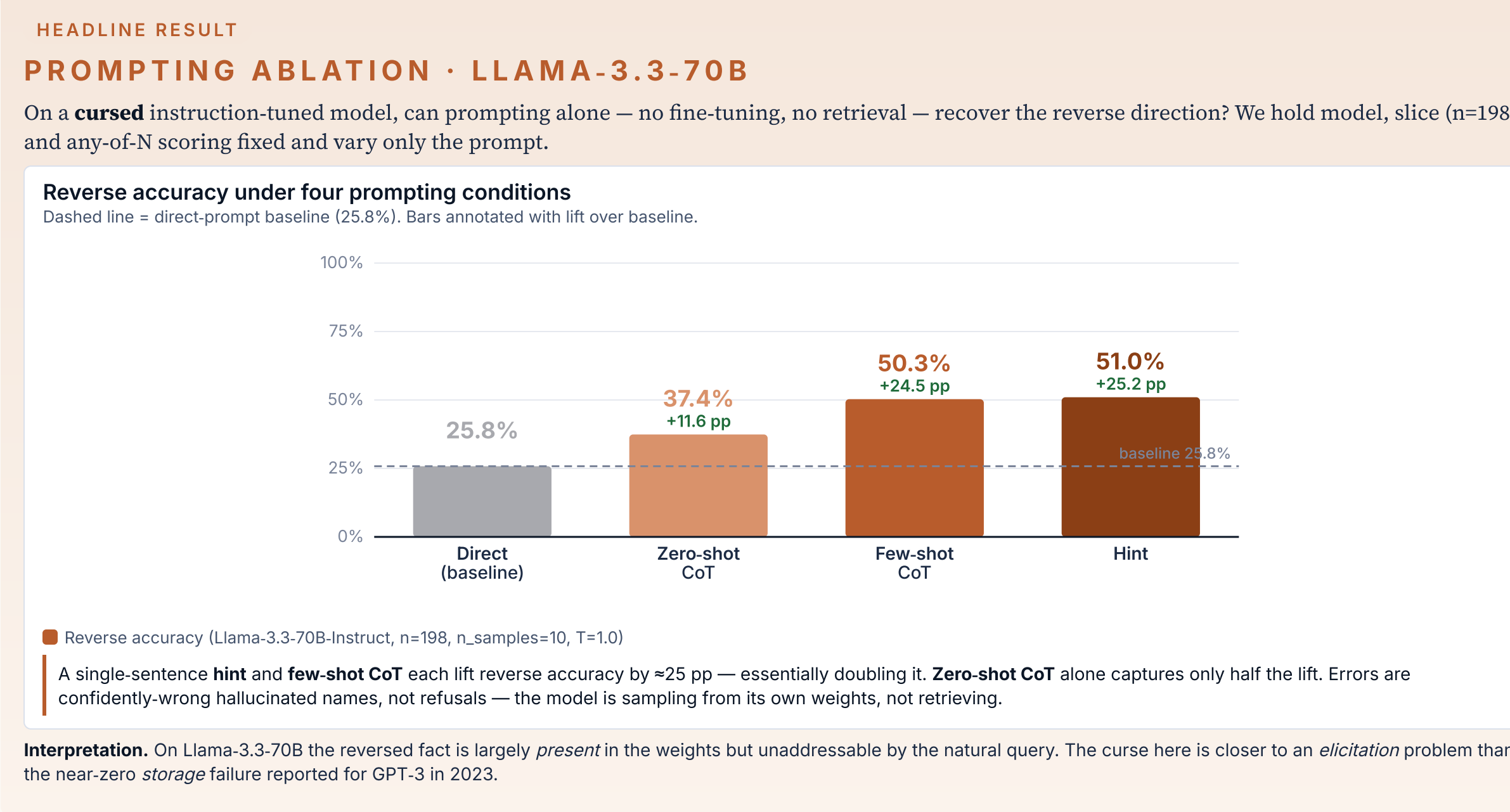
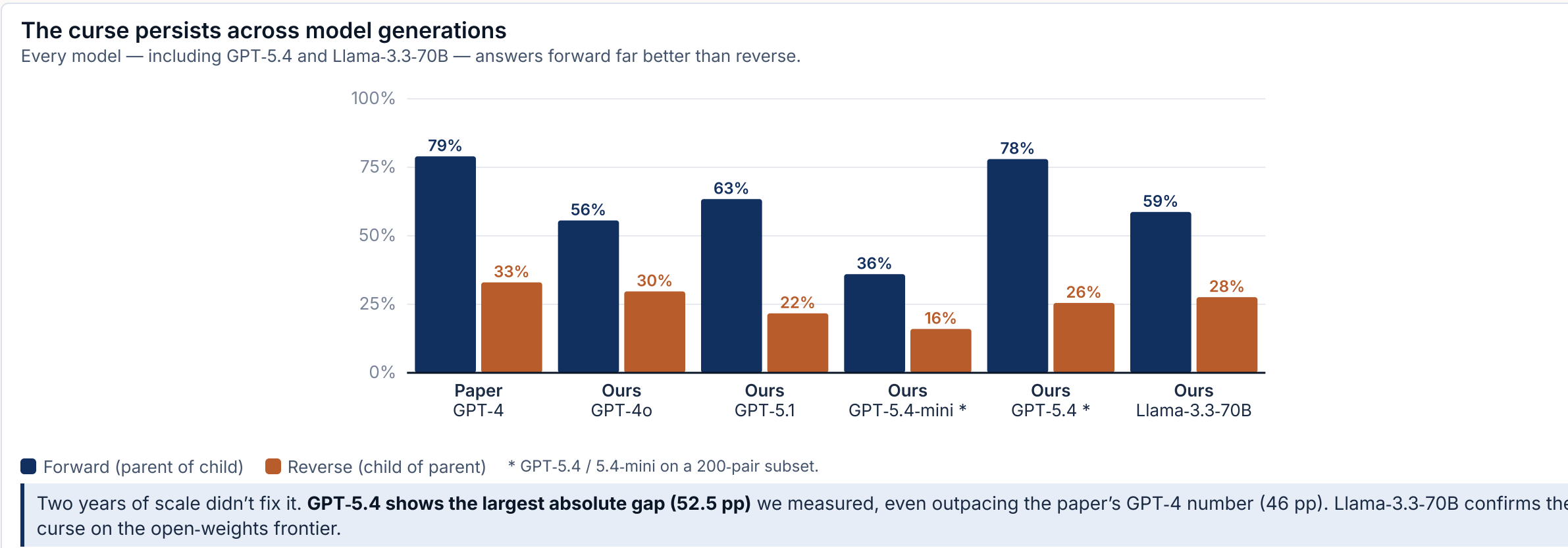
EXPERIMENT 1 · SYNTHETIC BIOGRAPHIES

Fine-tune on 300 fictitious facts in a fixed direction (D2P = description→name, P2D = name→description), then test exact-match accuracy in both orders.



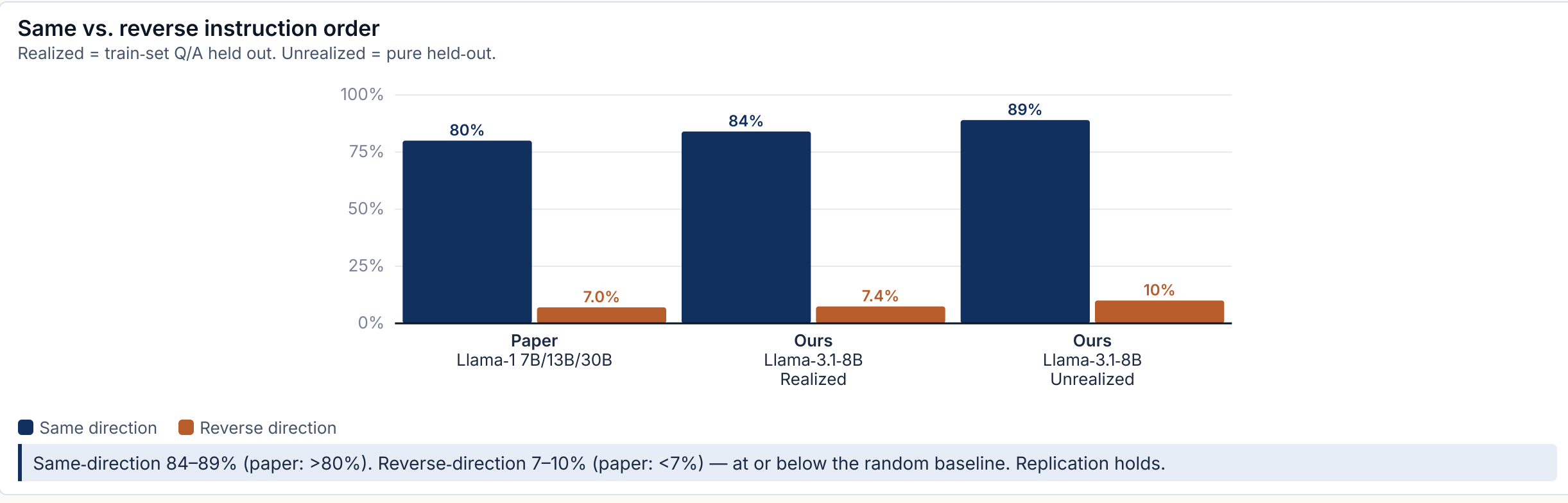
EXPERIMENT 2 · REAL CELEBRITIES, NO FINE-TUNING

Inference-only test on 1,513 child↔parent pairs from the paper’s celebrity set. Same any-of-N scoring, same few-shot prompt as Berglund 2024 (App. C.1). Forward = “*Who is X’s parent?*” Reverse = “*Name a child of Y.*”



EXPERIMENT 3 · INSTRUCTION REVERSAL

Fine-tune on instructions of the form “Answer <Q> with <A>” (or its reverse), then test on held-out Q/A. Replicates Berglund 2024 §2.3 on Llama-3.1-8B (LoRA, 20 epochs).



IN PROGRESS · TWO NEW BASELINES

PENDING A · Multihop Reversal

Train on forward-only chains A→B→C(→D), test reverse at 1, 2 and 3 hops. Does the deficit *compound* with hop-count, or plateau at chance from hop 1?

CONDITION	FORWARD	REVERSE	GAP
1-hop (Exp 1)	91.7%	9.3%	82.4 pp
2-hop	—	—	—
3-hop	—	—	—

PENDING B · Bidirectional Training Sweep

For $k \in \{0, 10, 25, 50, 75, 100\}$ % of training pairs, append the reversed sentence. How much bidirectional exposure is needed to close the gap — linear in k , or threshold-shaped?

k	Gap
k=0	9.3% (Exp 1)
k=100	upper bound

Curve to be filled. Baseline (k=0) and upper bound (k=100) anchor the y-axis.

CONCLUSIONS

- **The curse is robust.** Scale and instruction tuning have not eliminated the forward/reverse asymmetry on synthetic or real-world facts.
- **Open weights aren’t spared.** Llama-3.3-70B shows a 31 pp gap on the celebrity test, comparable to GPT-4o and GPT-5.1.
- **Prompting helps — a lot.** A hint or a few CoT demos recovers most of the reverse direction on Llama-3.3-70B without touching the weights.
- **Storage vs. elicitation.** The curse on modern models looks less like missing knowledge and more like a directional addressing failure.

